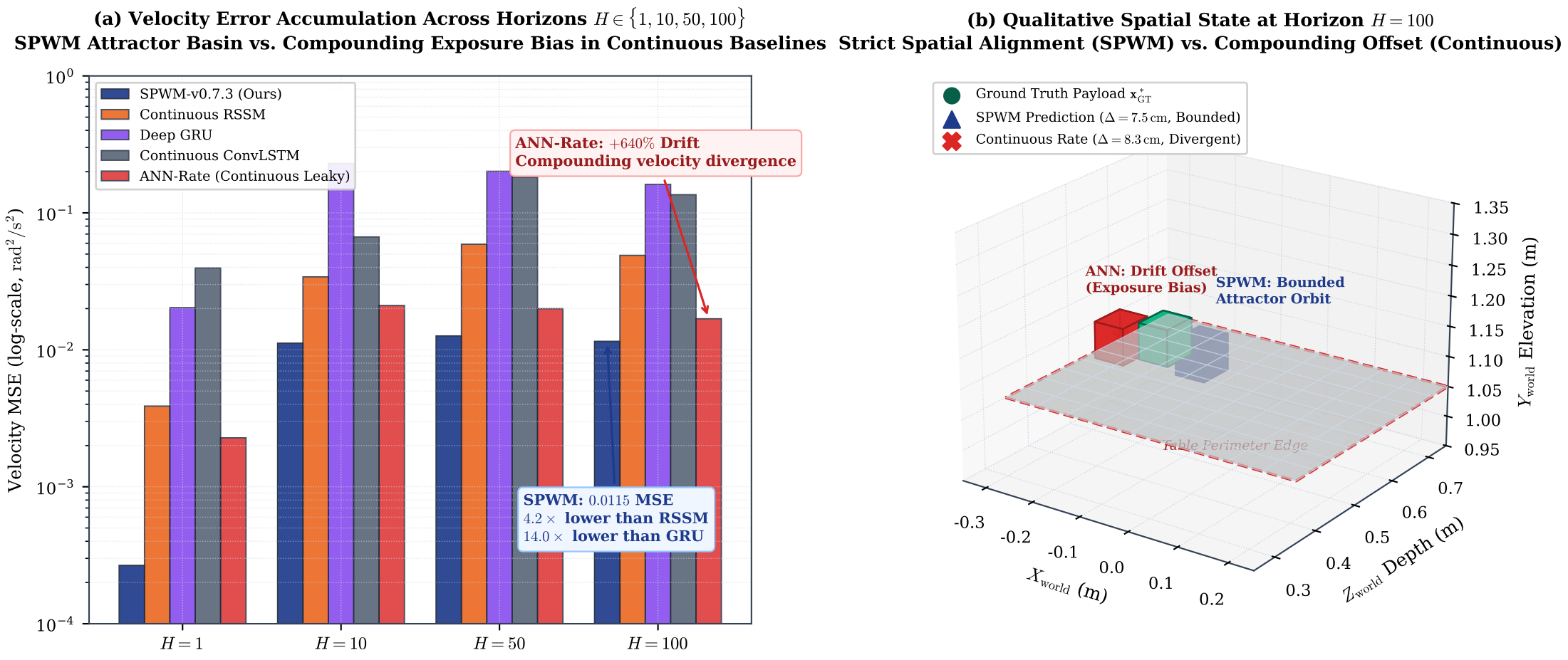

(a) Velocity Error Accumulation Across Horizons $H \in \{1, 10, 50, 100\}$
SPWM Attractor Basin vs. Compounding Exposure Bias in Continuous Baselines
SPWM-v0.7.3 (Ours)
Continuous RSSM
Deep GRU
Continuous ConvLSTM
ANN-Rate (Continuous Leaky)
ANN-Rate: +640% Drift
Compounding velocity divergence
SPWM: 0.0115 MSE
4.2 × lower than RSSM
14.0 × lower than GRU
Velocity MSE (log-scale, rad²/s²)
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
$H = 1$
$H = 10$
$H = 50$
$H = 100$
(b) Qualitative Spatial State at Horizon $H = 100$
Strict Spatial Alignment (SPWM) vs. Compounding Offset (Continuous)
Ground Truth Payload $\mathbf{x}^{*}_{GT}$
SPWM Prediction ($\Delta = 7.5$ cm, Bounded)
Continuous Rate ($\Delta = 8.3$ cm, Divergent)
ANN: Drift Offset
(Exposure Bias)
SPWM: Bounded
Attractor Orbit
Table Perimeter Edge
X_{world} (m)
-0.3
-0.2
-0.1
0.0
0.1
0.2
Z_{world} Depth (m)
0.3
0.4
0.5
0.6
0.7
Y_{world} Elevation (m)
1.35
1.30
1.25
1.20
1.15
1.10
1.05
1.00
0.95